EXP.NO-12	Implementing Security Features and Governance / Administration Practices in Zoho Creator

AIM:

To configure and implement security policies, role-based access control (RBAC), IP restrictions, and administrative governance practices within a Zoho Creator application to ensure data protection, compliance, and controlled access.

THEORY:

Application security and governance form the core foundation of enterprise-grade low-code solutions. Zoho Creator provides integrated administrative tools powered by Zoho Directory to manage system access and data integrity:

1. **Role-Based Access Control (RBAC):**
   a. Concept: Assigns specific roles (such as Administrator, Manager, or Employee) to users, restricting their access to designated forms, reports, and workflows.
   b. Benefit: Limits data exposure, enforces the principle of least privilege, and reduces the risk of internal security breaches.
2. **Security Policies (Password, MFA, & Sessions):**
   a. Concept: Governs authentication requirements by enforcing complex password rules, Multi-Factor Authentication (MFA), and session time-outs.
   b. Benefit: Prevents unauthorized account access and protects corporate data from credential theft.
3. **IP Restrictions:**
   a. Concept: Restricts user sign-ins and application access exclusively to pre-approved IP addresses or trusted corporate ranges.
   b. Benefit: Blocks intrusion attempts originating from untrusted or unknown networks.
4. **Audit Trails:**
   a. Concept: Maintains secure, tamper-proof logs capturing user activities, data alterations, and timestamp records.
   b. Benefit: Ensures user accountability and aids in regulatory compliance and troubleshooting.

PROCEDURE:

Step 1: Setting up Role-Based Access Control (RBAC)

1. Open your application in the Zoho Creator Developer Dashboard.
2. Navigate to Users and Control / Permissions under the settings menu.

3. Define distinct user roles (e.g., Sales_Agent, Branch_Manager, Admin).
4. Assign granular permissions to each role, specifying whether users can Add, View, Edit, or Delete records across specific forms and custom reports.

Step 2: Configuring Security Policies (Password and MFA)

1. Go to the Governance section on the left-side pane of your Zoho Creator/Zoho Directory dashboard.
2. Click on Security Policies and create a new policy profile.
3. Configure the Password Policy parameters (minimum length, special characters, digit requirement, and expiry duration).
4. Enable Multi-Factor Authentication (MFA), selecting supported modes such as Zoho OneAuth, Google Authenticator, or hardware keys (YubiKey).

Step 3: Enforcing IP Restrictions

1. Within the active Security Policy settings, navigate to the Allowed IPs section.
2. Specify the organization's trusted network IP addresses or subnet ranges.
3. Save the policy. Any sign-in request attempted outside these configured network boundaries will be automatically blocked.

Step 4: Monitoring Audit Trails and Session Controls

1. Under Governance / Administration, access the Audit Trails module to check chronological records of user activities.
2. Configure Advanced Session Settings (such as Session Lifetime and Idle Session Timeout) to automatically sign out inactive users.

RESULT:

Security features and governance controls were successfully configured in the Zoho Creator application. Role-based access control secured data visibility based on user responsibilities, security policies and MFA enforced stringent authentication, and IP restrictions successfully safeguarded the application against unauthorized external access attempts.